Introduction to Probabilistic Numerics

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What do economists compute?

- Approximations of unknown quality – true solutions could be arbitrarily different... What can we do?
    - Change model
    - Hope to make probabilistic statements about the quality of a solution: "With high probability, x is the maximum of f(.)"

Early views

Ainsi la probabilité pour que la sixième décimale d'un nombre dans une table de logarithmes soit égale à 6 est a priori de 1/10; en réalité, toutes les données du problème sont bien déterminées, et, si nous voulions nous en donner la peine, nous connaîtrions exactement cette probabilité.

H. Poincaré. Calcul des Probabilités. 1896.

Early views

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It may be too much effort to obtain the exact answer. Need an algorithm that allows us to narrow the probability even if it stays interior.

Probabilistic Numerics in the 21st Century

- C. Rasmusson, C Williams: Gaussian Processes for Machine Learning, 2006.
- P. Hennig, M. Osborne, H. Kersting: Probabilistic Numerics, 2022.
- A. Krause, J. Hübotter: Probabilistic Artificial Intelligence, 2025.
- R. Garnett: Bayesian Optimization, 2023.

Probabilistic Comp. Econ.

- Consider a generic model $\mathscr{M}(\cdot)$ which has N (uncertain) parameters $\theta \in \Theta \subset \mathbb{R}^N$ as an input and model outputs $\mathscr{M}(\theta) \in \mathbb{R}^k$. Take $\widehat{\mathscr{M}}(\cdot)$ to be a probabilistic surrogate (e.g. Gaussian process)
- M computational experiments are conducted at $\theta_1, \ldots, \theta_M \in \Theta$, return $y_1, \ldots, y_M$ with $y_i = \mathscr{M}(\theta_i)$
- Examine posterior distribution of $\widehat{\mathscr{M}}(\cdot)$ given $(\theta_i, y_i)_{i=1}^M$

Gaussian processes

- A Gaussian process is a collection of random variables, indexed by $x \in \mathbb{R}^n$, any finite number of which have a joint Gaussian distribution
- Completely specified by its mean function m(x) and covariance function k(x,x')

Covariance function

- Covariance function is a (positive semi-definite and symmetric) kernel k(x,x')=k(x',x),

$$\sum_{i=1}^{n}\sum_{j=1}^{n} k(x_i, x_j) c_i c_j \geq 0$$

- Often want kernel to be stationary k(x,x')=f(x-x'), smooth, isotropic k(x,x')=f(|x-x'|)
- Sum of kernels are kernels, certain product of kernels are kernels... Infinite possibilities to waste time

Gaussian processes

$$g(\,.\,) \sim \mathcal{GP}(m(\,.\,), k(\,.\,,.\,)) \Rightarrow E(g(x)) = m(x),\ cov(g(x,), g(x')) = k(x, x')$$

$$k_{SE}(x, x') = \sigma_f^2 exp\left(-\frac{1}{2}\sum_{d=1}^{D}\frac{(x_d - x_d')^2}{\ell_d^2}\right)$$

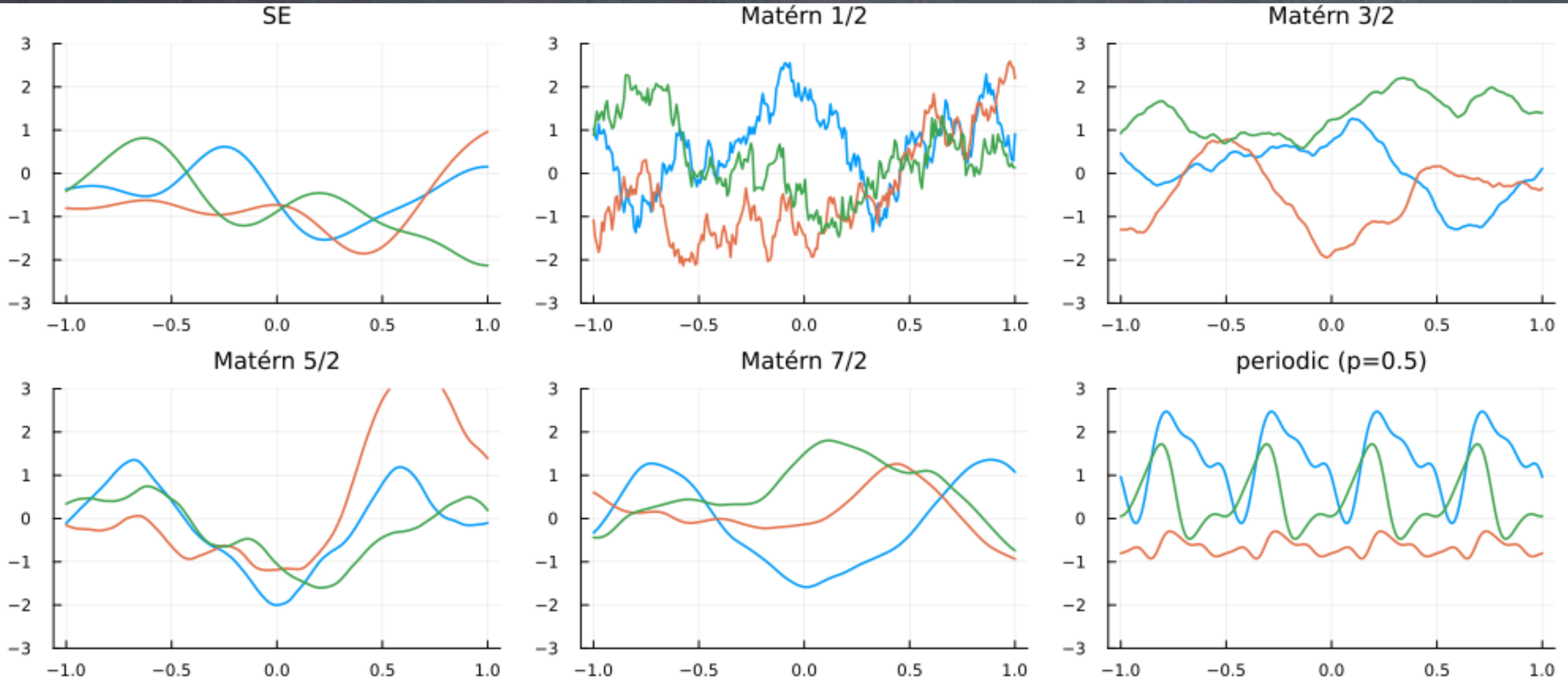
Hyperparameter

Matern kernel

$$k_{1/2}(x, x') = \sigma_S \exp\left(-\frac{r}{\rho}\right), \; r = |x - x'|$$

$$k_{3/2}(x, x') = \sigma_S \left(1 + \frac{\sqrt{3}\, r}{\rho}\right) \exp\left(-\frac{\sqrt{3}\, r}{\rho}\right), \; r = |x - x'|$$

$$k_{5/2}(x, x') = \sigma_S \left(1 + \frac{\sqrt{5}\, r}{\rho} + \frac{5r^2}{3\rho^2}\right) \exp\left(-\frac{\sqrt{5}\, r}{\rho}\right), \; r = |x - x'|$$


SE
Matérn 1/2
Matérn 3/2
Matérn 5/2
Matérn 7/2
periodic (p=0.5)
3
2
1
0
−1
−2
−3
−1.0
−0.5
0.0
0.5
1.0

Conditional normals

- Joint distribution:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right)$$

- Conditional distribution:

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

GP regression

Observations $(x^i, y^i)_{i=1}^n, \quad y^i = f(x^i) + \epsilon^i$

iid normal with std s

$$X = (x^1, \ldots, x^n),\ K(x, X) = (k(x, x^1), \ldots, k(x, x^n))$$

$K = (k(x^i, x^j))_{ij}$ Prior mean=0

GP regression

$$\mathcal{D}_n = \{(x_i, y_i)\}_{i=1}^n$$

$$g(x) \mid \mathcal{D}_n \sim \mathcal{N}(\mu_n(x), \sigma_n^2(x))$$

$$\mu_n(x) = K(x, X)^\top (K + s^2 I)^{-1} y$$

—> radial basis function interpolation

$$\sigma_n^2(x) = k(x, x) - K(x, X)^\top (K + s^2 I)^{-1} K(x, X)$$

Probabilistic numerics

Assume that f(.) is a sample path of the Gaussian process

$$\mathbb{P}(f(x) \geq \xi) = \Phi\left(\frac{\mu_n(x) - \xi}{\sigma_n(x)}\right)$$

Intuition (R-W, page 3)

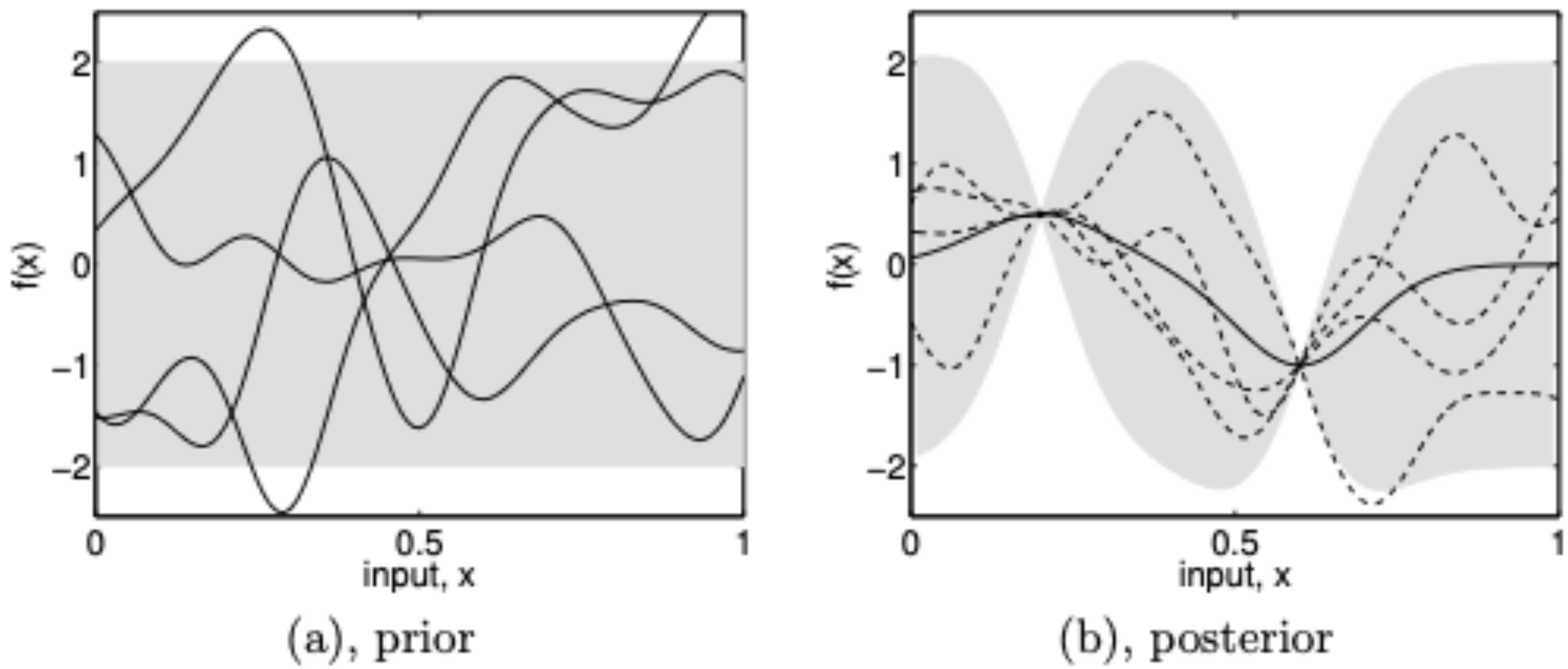


Figure 1.1: Panel (a) shows four samples drawn from the prior distribution. Panel (b) shows the situation after two datapoints have been observed. The mean prediction is shown as the solid line and four samples from the posterior are shown as dashed lines. In both plots the shaded region denotes twice the standard deviation at each input value x.

Reproducing kernel Hilbert space

- Let $\mathscr{H}$ be a Hilbert space of real-valued functions on a set $\mathscr{X}$ with inner product $\langle \cdot, \cdot \rangle_{\mathscr{H}}$. We say $\mathscr{H}$ is a reproducing kernel Hilbert space (RKHS) if there exists a function $k : \mathscr{X} \times \mathscr{X} \to \mathbb{R}$ such that:
  - For all $x \in \mathscr{X}$, $k(\cdot, x) \in \mathscr{H}$
  - For all $f \in \mathscr{H}$ and $x \in \mathscr{X}$, $f(x) = \langle f, k(\cdot, x) \rangle_{\mathscr{H}}$
- The function k is called the reproducing kernel of $\mathscr{H}$.

Is f a sample path?

- Karvonen (2023): A C^∞ function f is in the sample support set if and only if its Hermite-function expansion coefficients $\{c_n\}$ satisfy: there exist constants $0 < c \leq C < \infty$ such that $c \leq |c_n|^2/\lambda_n \leq C$ for all n, where $\{\lambda_n\}$ are the (super-exponentially decaying) eigenvalues of the SE kernel's integral operator…
- GP with SE kernel: van der Vaart and van Zanten (2011): the probability of a draw f falling in a neighbourhood of a given continuous f is positive, no matter how small the neighbourhood.

Good Approximation

Van der Vaart and van Zanten (2008):

The probability of a draw of the Gaussian process falling in a neighbourhood of a given continuous $ f $ is typically positive, no matter how small the neighbourhood. This means every continuous function in the interior of the sup-norm closure of the RKHS is in the support. For the SE kernel on a compact domain, the RKHS is dense in the space of continuous function so the support in the sup-norm topology is all of C(T) – meaning every continuous function has positive probability of being approached arbitrarily well.

Detour: RBF interpolation

- Linear approximation methods great in one dimension, but not for interpolation in high dimensions: Unfortunately Mairhuber-Curtis theorem kills it
- This implies that the basis functions must depend on the points: RBF an example
- Can interpolate any points and is a universal approximation
- Optimal recovery properties similar to univariate splines

Detour (cont)

- Take function f in RKHS of k, assume N observations. Unique interpolating RBF with kernel k.

- Consider all possible linear recovery schemes: Take any functions $u_j(\,.\,)$ and define $f_u(\,.\,) := \sum_{j=1}^{N} u_j(\,.\,)f(x_j)$. Then

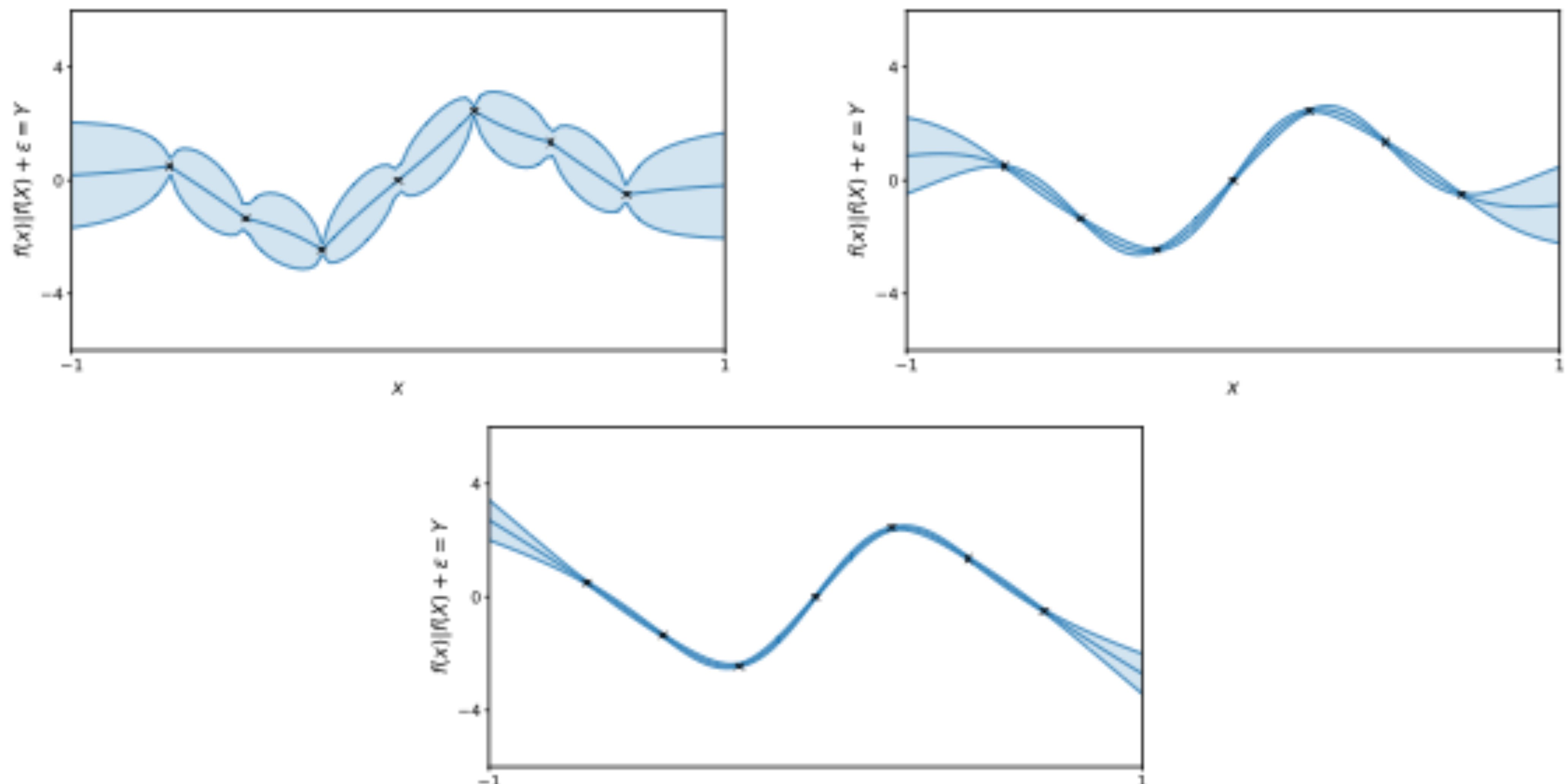
$$\inf_u \sup_{\|f\|_{\mathcal{K}} \le 1} |f(x) - f_u(x)| = \sup_{\|f\|_{\mathcal{K}} \le 1} |f(x) - f^*(x)|$$

The kernel problem

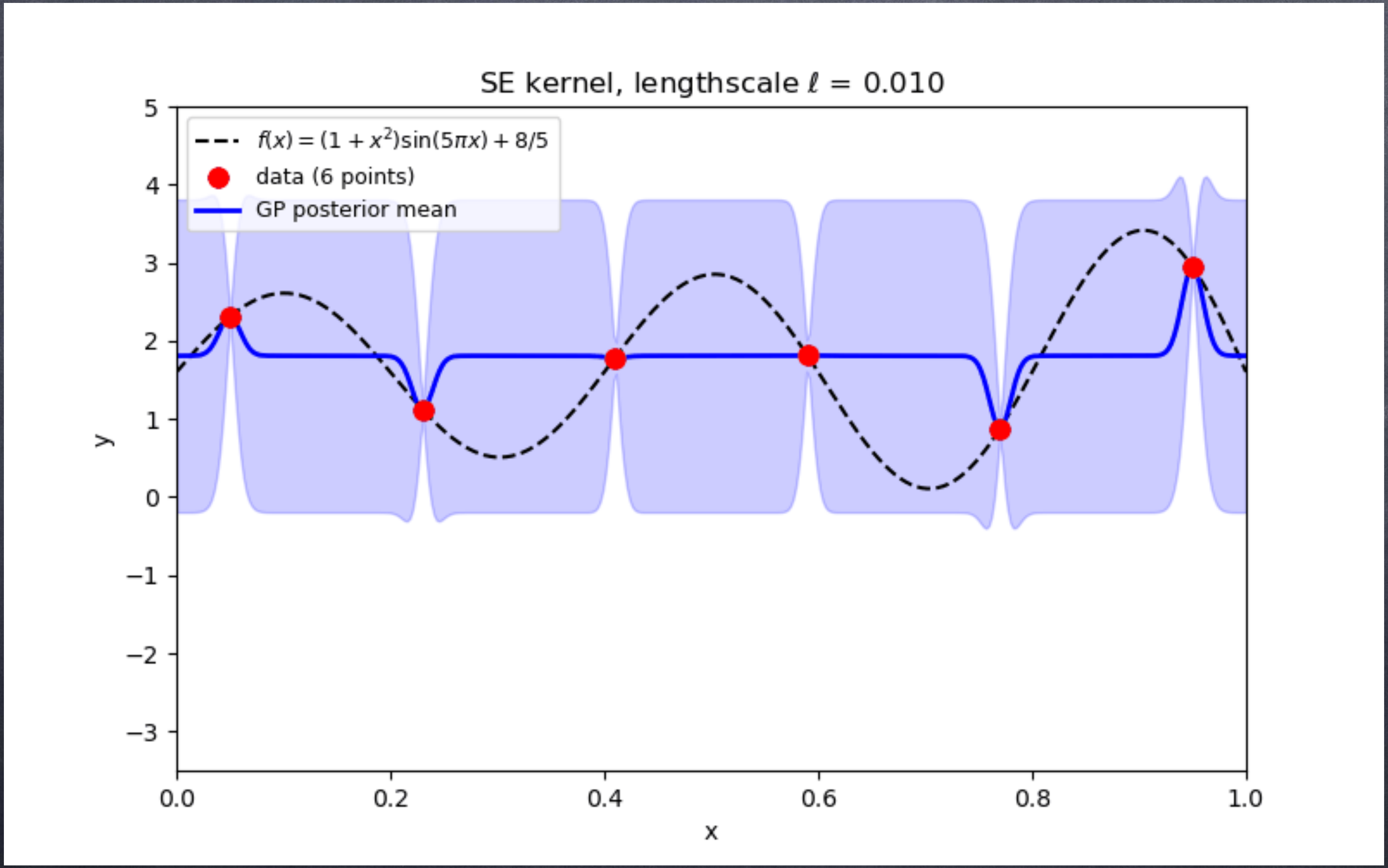
- To assume true function is a sample path of a GP we need to characterize set of these functions
  - What are the restrictions on f ?
  - Degree of smoothness important
  - If you know that f is C^{∞}, use SE kernel
- What if kernel is misspecified?

Large effect?

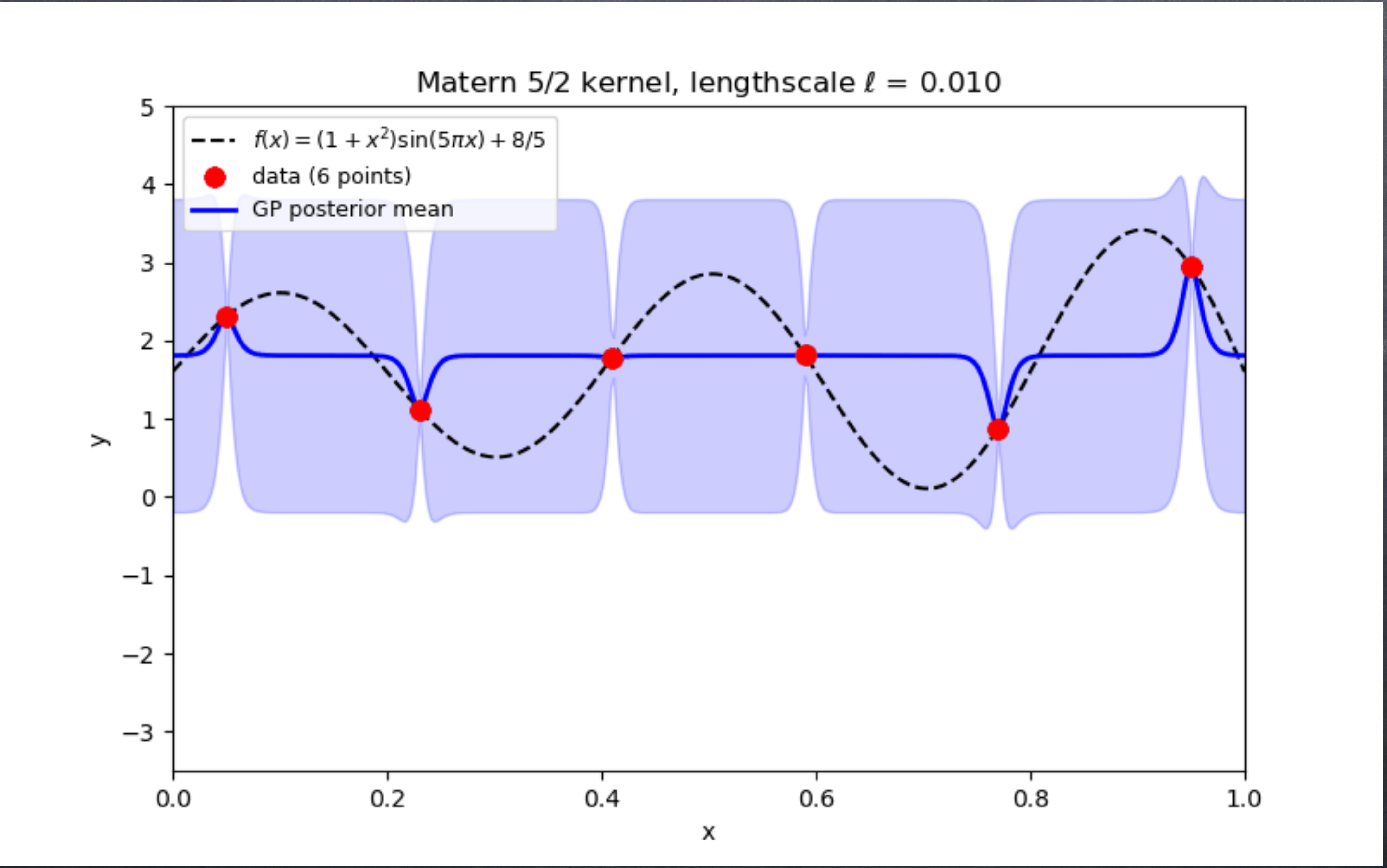
The kernel has a huge impact on the model: (Matérn 1/2, Matérn 5/2 and Arccosine kernels)



Role of hyperparameters



Different kernel..



Estimating hyperparameter

- Marginalising over the latent values gives a
- Gaussian distribution over the observed targets, and its log density— the log marginal likelihood
  $$\log p(y \mid X, \theta) = -\frac{1}{2}\, y^\top K_y^{-1} y \;-\; \frac{1}{2}\log|K_y| \;-\; \frac{n}{2}\log 2\pi\,.$$
- Maximize over theta

Applications

- Integration: Bayesian Quadrature
- Global Optimization: Bayesian Optimization

Bayesian Quadrature

- Sample (how) the true function, do GP regression and integrate the posterior mean
- For many kernels and densities, we have analytic expressions
- Recall Gaussian Quadrature!

Bayesian Quadrature

- Want $E\left(\int_x g_n(x)dF(x)\right) = \int_x \mu_n(x)dF(x)$
- $\mu_n(x) = K(x, X)^\top (K + s^2 I)^{-1} y = K(x, X)Cy$

Bayesian Quadrature

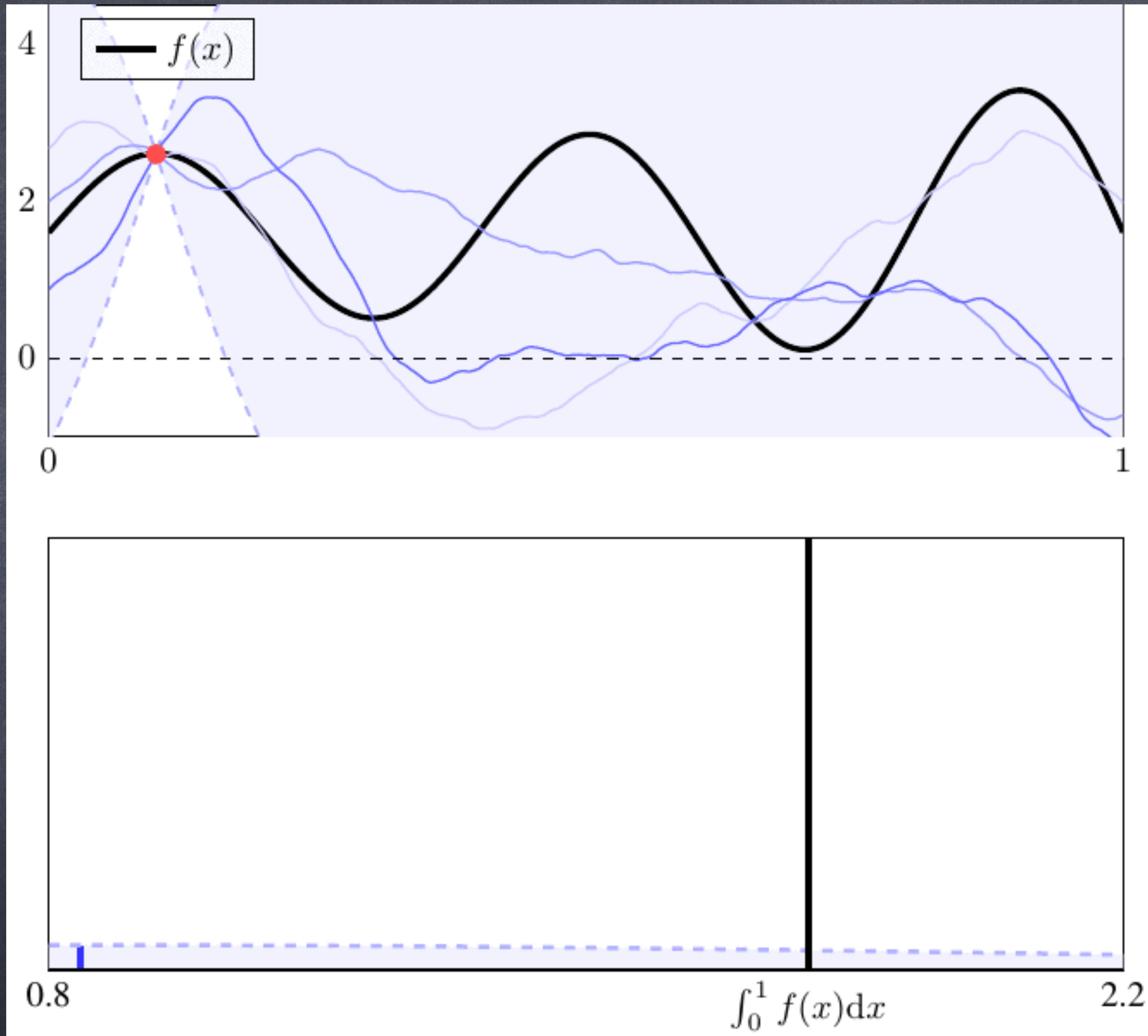
- Integrate with respect to cdf F. Let $g_n(x)$ be the GP after n observations
- $E\left(\int_x g_n(x)dF(x)\right) = \int_x k(x,X)dF(x)C^{-1}y$
- $var = \int_x \int_y k(x,y)dF(x)dF(y) - \int_x k(x,X)dF(x)C^{-1}\int_x k(X,x)dF(x)$
- $\mathbb{P}\left(|E\int_x g_n(x)dF(x) - \int_x g_n(x)dF(x)| \leq \sqrt{var}F_{\mathcal{N}}^{-1}(1-\frac{a}{2})\right) = 1-a$

Example

- Consider $\int_0^1 f(x)\mathrm{d}x$
- Approximate f by a zero-mean Gaussian process prior with the Matérn covariance function of smoothness 3/2: $k(x,y) = (1+\sqrt{3}\,|x-y|/\rho)\exp(-\sqrt{3}\,|x-y|/\rho)$
- Tedious but not rocket science:

$$\int_0^1 k(y,x)\mathrm{d}y = \frac{4\rho}{\sqrt{3}} - \frac{1}{3}\exp\left(\frac{\sqrt{3}(x-1)}{\rho}\right)(3+2\sqrt{3}\rho-3x) - \frac{1}{3}\exp\left(-\frac{\sqrt{3}x}{\rho}\right)(3x+2\sqrt{3}\rho)$$

$$\int_0^1\int_0^1 k(x,y)\mathrm{d}x\ \mathrm{d}y = \frac{2\rho}{3}\left[2\sqrt{3}-3\rho+\exp\left(-\frac{\sqrt{3}}{\rho}\right)(\sqrt{3}+3\rho)\right]$$



$f(x)$
4
2
0
0
1
0.8
$\int_0^1 f(x)\mathrm{d}x$
2.2

Known integrals

$\mathcal{X}$	ν	k	Reference
$[0,1]^d$	Unif($\mathcal{X}$)	Wendland TP	Oates et al. (2019a)
$[0,1]^d$	Unif($\mathcal{X}$)	Matérn Weighted TP	Briol et al. (2019)
$[0,1]^d$	Unif($\mathcal{X}$)	Gaussian	Use of error function
$\mathbb{R}^d$	Mix. of Gaussians	Gaussian	Kennedy (1998)
$\mathbb{S}^d$	Unif($\mathcal{X}$)	Gegenbauer	Briol et al. (2019)
Arbitrary	Unif($\mathcal{X}$) or mix. of Gauss.	Trigonometric	Integration by parts
Arbitrary	Unif($\mathcal{X}$)	Splines	Wahba (1990)
Arbitrary	Known moments	Polynomial TP	Briol et al. (2015)
Arbitrary	Known $\partial \log \nu(x)$	Gradient-based kernel	Oates, Girolami, and Chopin (2017); Oates et al. (2019a)

Sampling

- Optimal design rule selects $X = [x_1, \ldots, x_N]$ to minimize variance of the integral. Can be computationally impossible.
- Sample instead from the N-determinantal point process of k
- Does not depend on y!

Adaptive Selection

- Sample sequentially where the variance of the integrand (not the integral) is large.

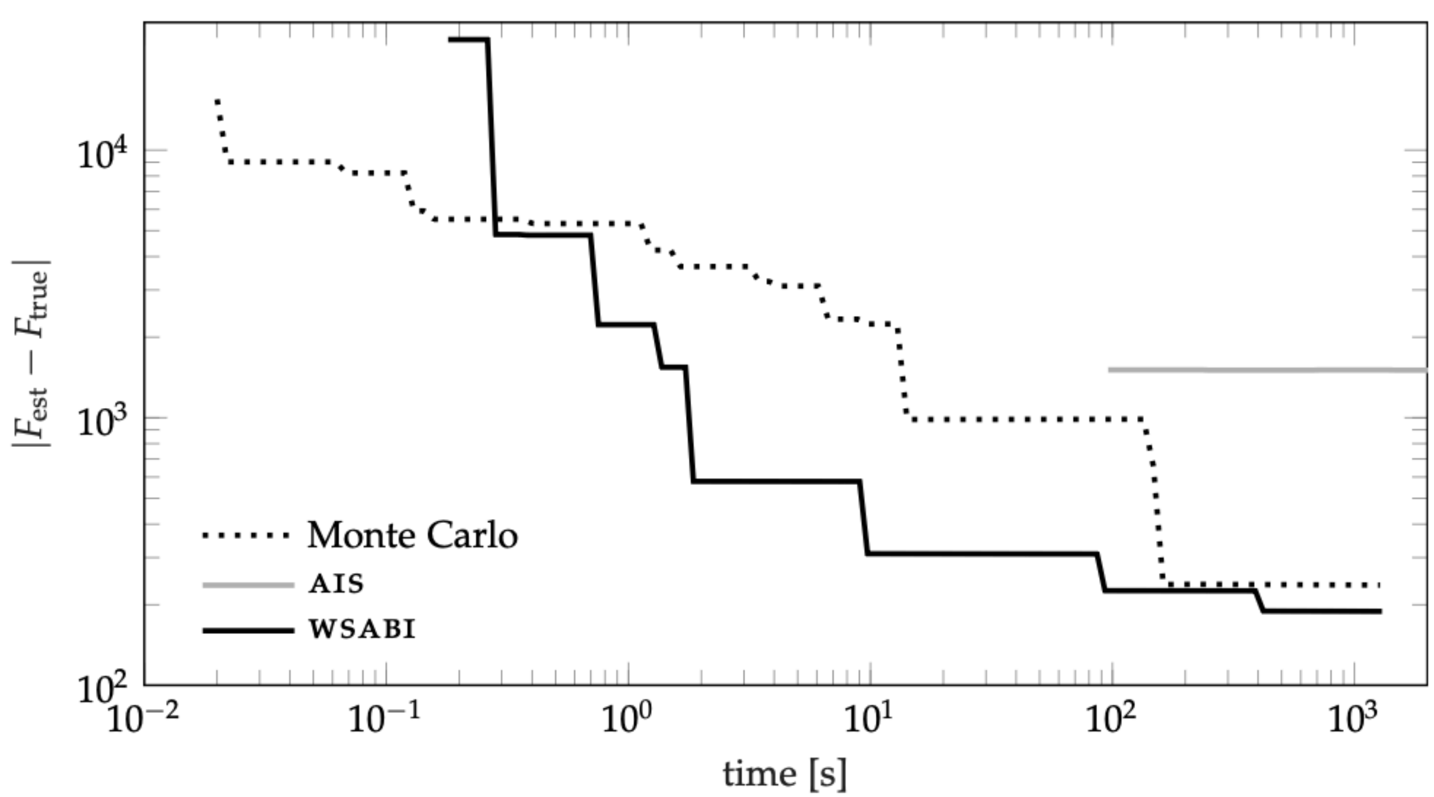


Figure 10.1: A comparison of approaches to estimating the marginal likelihood F from Eq. (10.11) for a GP regressor fitted to real data – the yacht hydrodynamics benchmark data set (Gerritsma, Onnink, and Versluis, 1981). The required integral was eight-dimensional. This plot is adapted from Gunter et al. (2014), which contains full details.

Sequential sampling

- Given N-1 observations, pick the Nth observation at the point where posterior std is maximized
- —> Bayesian optimization

BO - setup

- Consider the problem $x^* = \arg\max_{x \in \mathcal{X}} f(x)$
- A probabilistic model of the objective function -> GP surrogate
- An acquisition function that uses the surrogate model to determine the next point to evaluate: $x_{n+1} = \arg\max_{x \in \mathcal{X}} \alpha(x \mid \mathcal{D}_n)$

Some acquisition functions

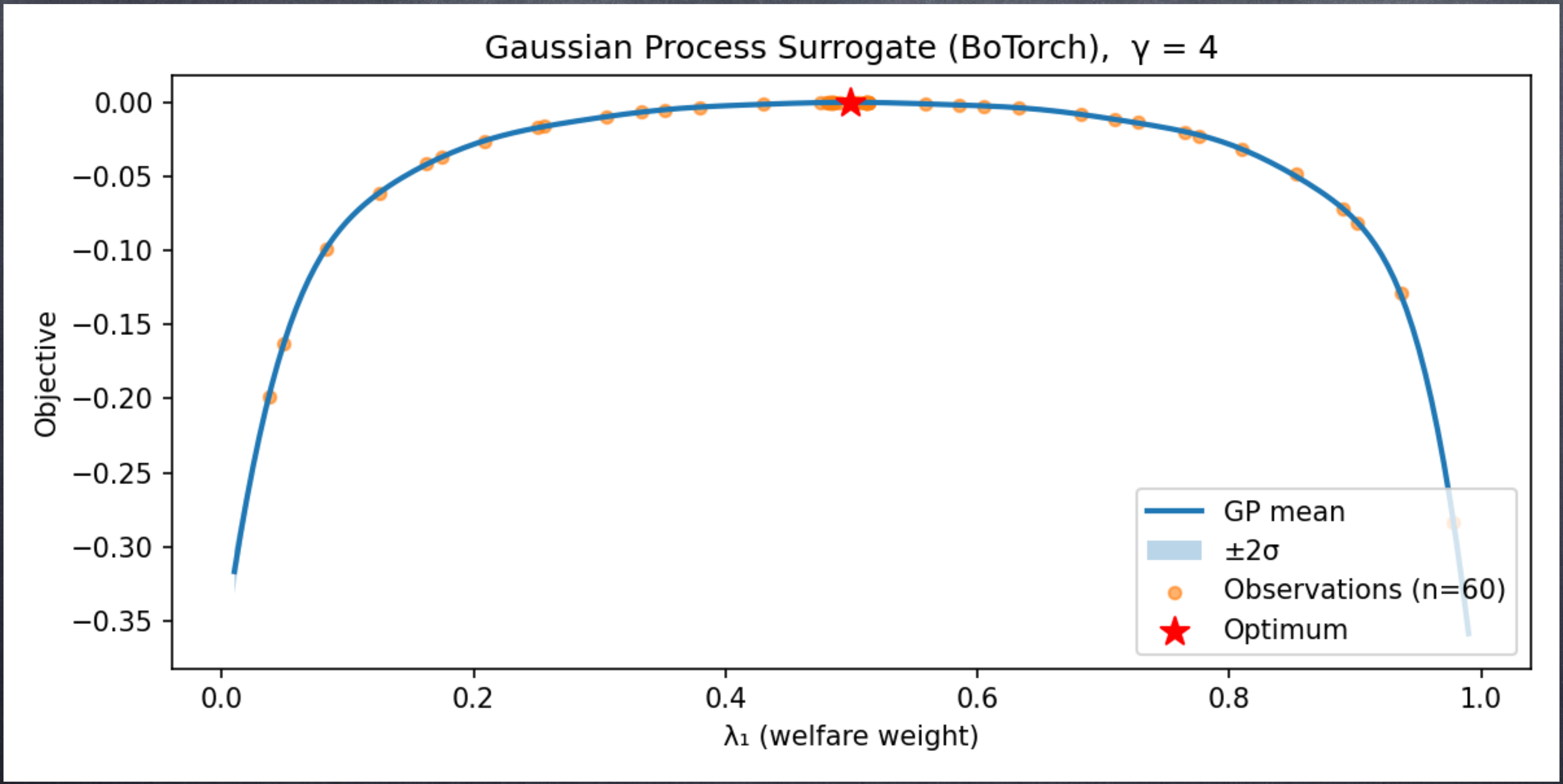
- PI:

$$\alpha(x \mid \mathscr{D}_n) = \ P(f(x) \geq f^+ + \xi) = \Phi\left(\frac{\mu_n(x) - f^+ - \xi}{\sigma_n(x)}\right)$$

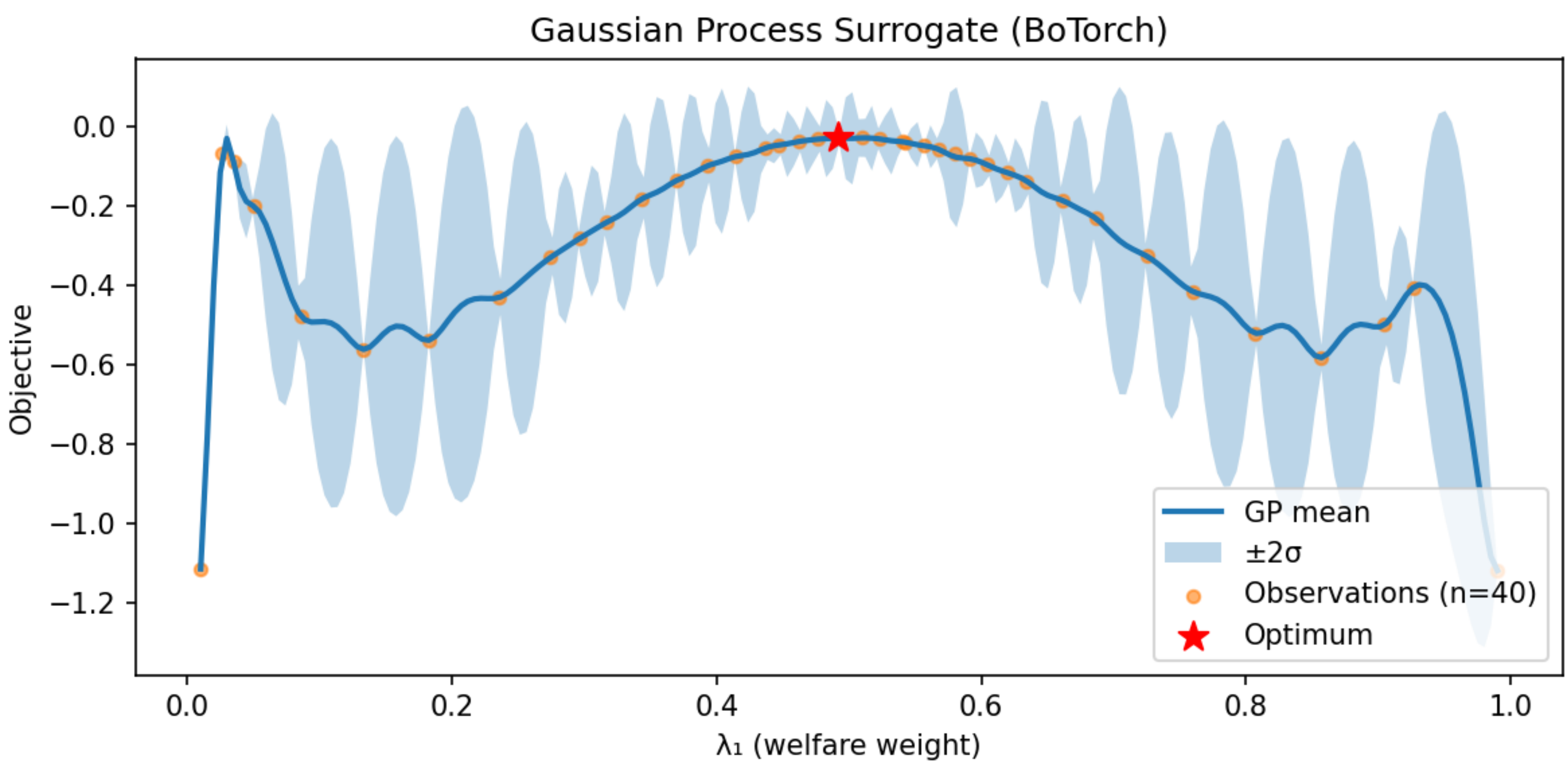
- UCB: $\alpha(x \mid \mathscr{D}_n) = \mu_n(x) + \beta\sigma_n(x)$

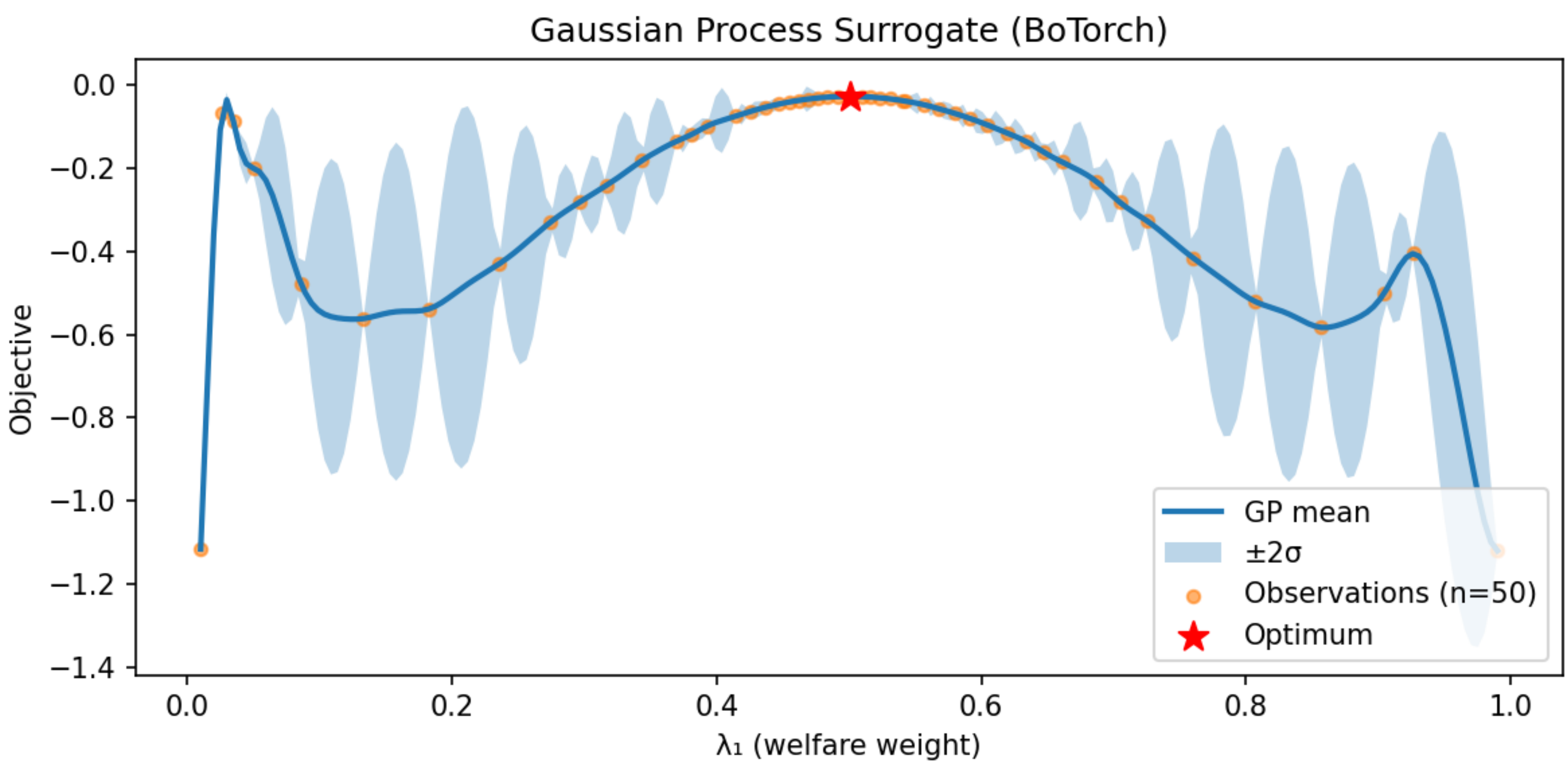
- EI (noiseless case)
$\alpha(x \mid \mathscr{D}_n) = E(\max_x(\mu(x) - f^*, 0))$

- Some Figures

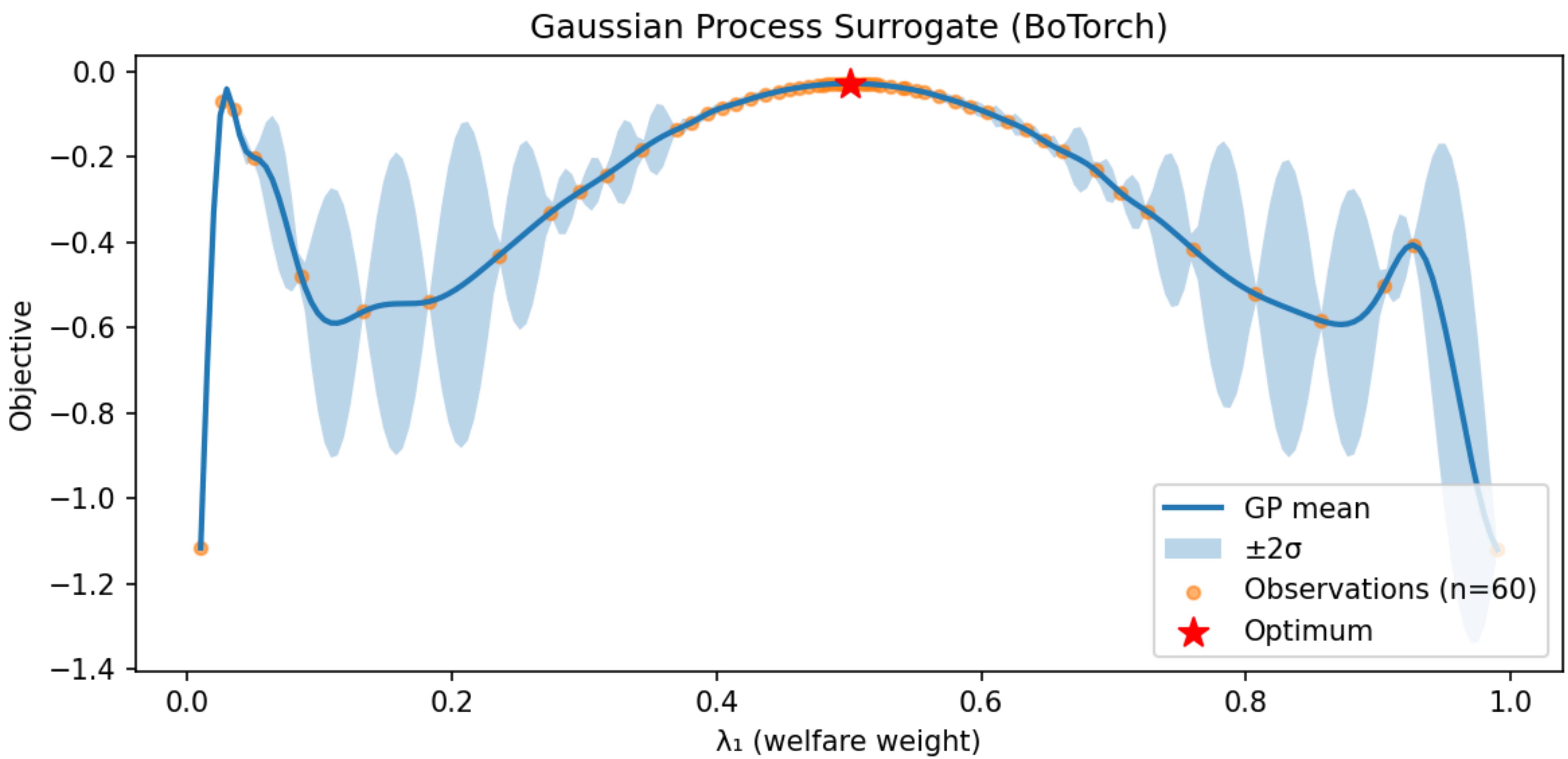


Gaussian Process Surrogate (BoTorch), γ = 4
Objective
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
−0.35
0.0
0.2
0.4
0.6
0.8
1.0
λ₁ (welfare weight)
GP mean
±2σ
Observations (n=60)
Optimum

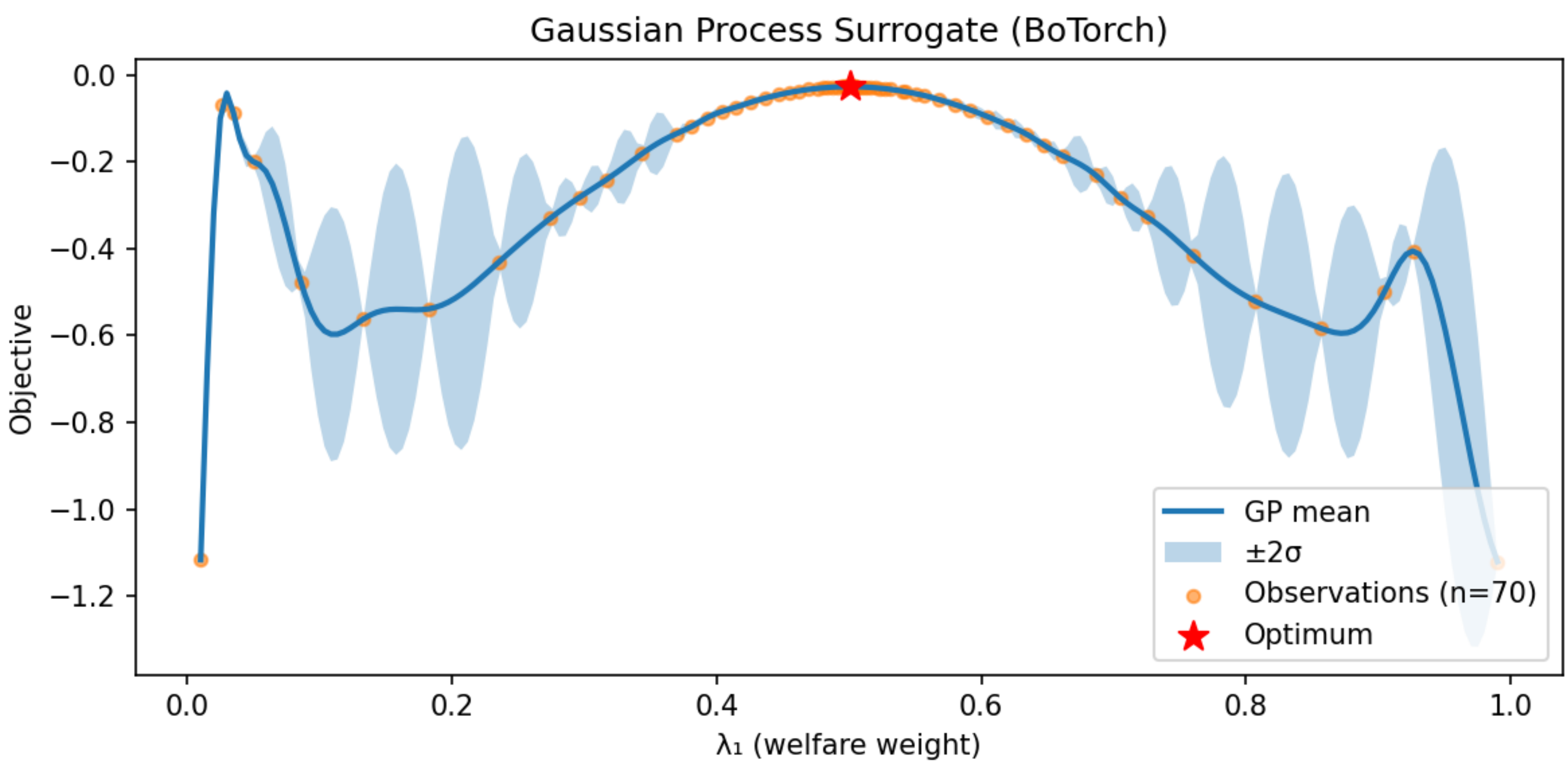

Gaussian Process Surrogate (BoTorch)
Objective
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
0.0
0.2
0.4
0.6
0.8
1.0
λ₁ (welfare weight)
GP mean
±2σ
Observations (n=40)
Optimum

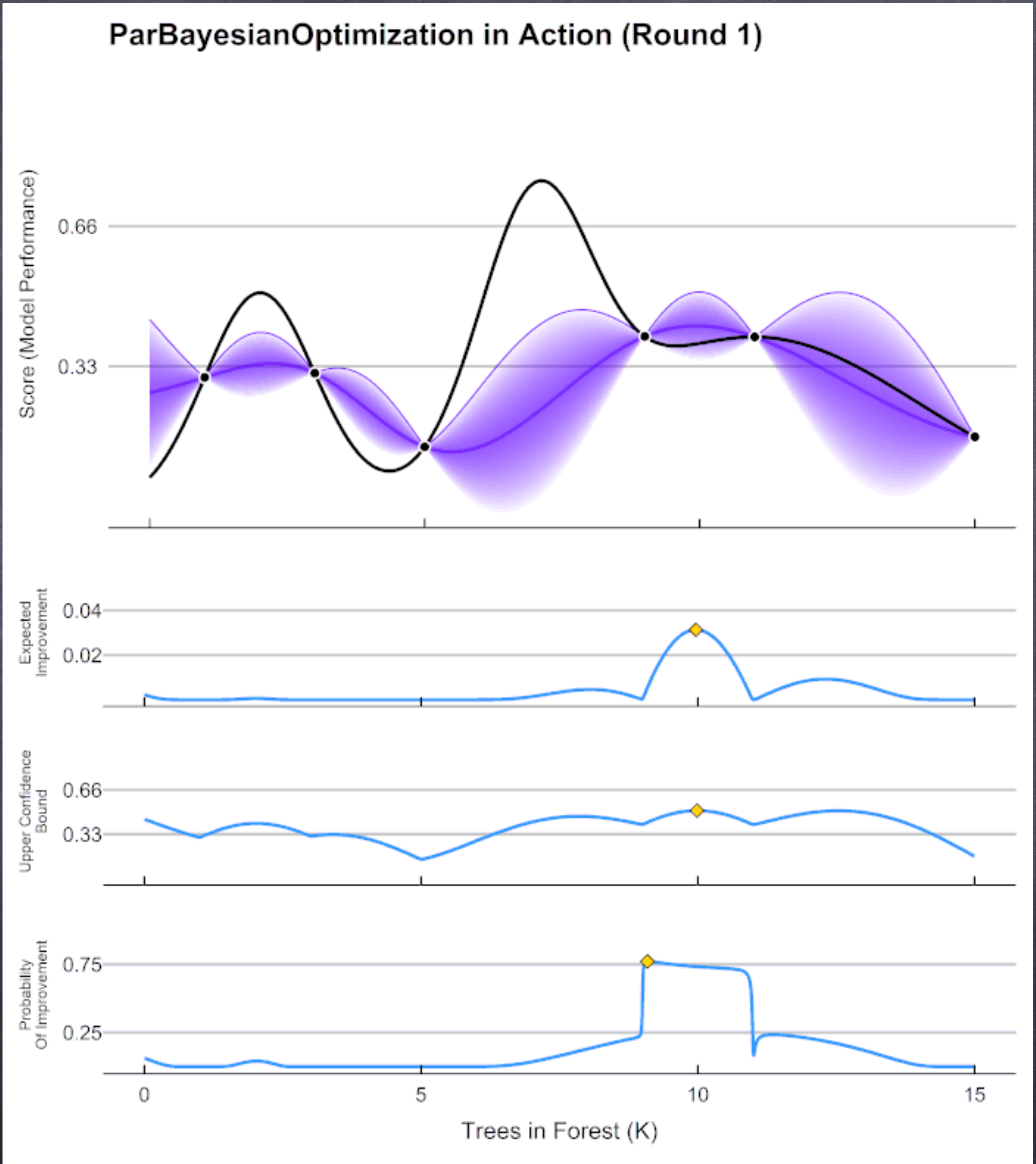


Gaussian Process Surrogate (BoTorch)
Objective
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
−1.4
0.0
0.2
0.4
0.6
0.8
1.0
λ₁ (welfare weight)
GP mean
±2σ
Observations (n=50)
Optimum



Gaussian Process Surrogate (BoTorch)
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
−1.4
Objective
0.0
0.2
0.4
0.6
0.8
1.0
λ₁ (welfare weight)
GP mean
±2σ
Observations (n=60)
Optimum


Gaussian Process Surrogate (BoTorch)
Objective
λ₁ (welfare weight)
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
0.0
0.2
0.4
0.6
0.8
1.0
GP mean
±2σ
Observations (n=70)
Optimum



https://en.wikipedia.org/wiki/en:Creative_Commons

BO - result

- Consider the problem $x^* = \arg\max_{x \in \mathcal{X}} f(x)$
- Find a $\tilde{x} \in \mathcal{X}$ such that for any x, $\mathbb{P}\left(f(x) \geq f(\tilde{x}) + \delta\right) \leq \epsilon$
- A better statement is possible?
- Need a Lipschitz constant on f to obtain the promised result...
- Derivative of a GP (if differentiable) is a GP

Lipschitz bound on f

- Suppose the sample path of GP is C1
- Borell-TIS inequality:

$$\mathbb{P}\left(\max_i |d_i| > \lambda\right) \leq 4\exp\left(-\frac{\lambda^2}{b_i^2}\right),\ b_i = 2\sqrt{2}\,\mathbb{E}\max_i |d_i|$$

- $\mathbb{E}\sup_{x\in\mathcal{X}} f(x) \leq 12\sqrt{6d}\max\left(\max_{x\in\mathcal{X}}\sqrt{k(x,x)}, \sqrt{rL_k}\right)$

Probably optimal

- Find Lipschitz bound that holds with high probability (e.g. 0.99).
- Find $\tilde{x}, B(\tilde{x})$ such that $\mathbb{P}\left(\max_{x \notin B(\tilde{x})} f(x) \geq f(\tilde{x})\right) \leq \epsilon$
- Assumes unique local maximum that is significantly larger than other local maxima, often true in economics

Practical considerations

- Gaussian process regression and Bayesian optimization have attracted a rich ecosystem of software implementations.
- BoTorch is a state-of-the-art Python library, developed by Meta and built on PyTorch

Conclusion

- The core idea is to build a surrogate model of the objective function and use it to decide where to evaluate next.
- This surrogate must be a probabilistic model, such as a Gaussian process, as the probabilistic treatment of uncertainty is vital: one must be able to reason about the uncertainty in unvisited regions of the objective function's domain.